

Pablo Nuñez Martinez

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First-year M.S. Aeronautics and Astronautics student specializing in aerodynamics, CFD, and composite structures. Experienced in wind-tunnel-validated CFD design and the leadership of large, multidisciplinary engineering teams, including founding a university electric racing team. Focused on turning simulation into validated, manufacturable hardware.

EDUCATION

Stanford University

M.S., Aeronautics and Astronautics — GPA 3.88

Stanford, CA

Expected June 2027

Tec de Monterrey

B.S., Mechanical Engineering — GPA 97.1/100

Guadalajara, Mexico

June 2025

McGill University

Study Abroad, Aerospace Engineering — GPA 4.0

Montréal, QC

Aug 2024 – Dec 2024

EXPERIENCE

Stanford Solar Car Project — World Solar Challenge

Aerodynamics Lead Engineer

Stanford, CA

Sept 2025 – Present

- Achieved a 65% drag reduction (drag coefficient of 0.00191 on wetted area) by iterating aerobody designs in ANSYS Fluent and Luminary Cloud, analyzing flow separation, pressure distribution, and wall shear stress with a focus on minimum drag and crosswind stability.
- Validated CFD methodology against historical wind-tunnel data, tuning turbulence, transition, and ground-modeling parameters, to reach >95% correlation between simulation and experiment.
- Built a parametric CAD model of the aerobody to accelerate design iteration and ensure compliance with competition regulations.
- Led a 10+ member aerodynamics subteam, managing tasks and timelines in Notion, training new members in CFD fundamentals, and maintaining documentation of research and design workflows.

ELYOS Racing Team — Shell Eco-marathon

Founder & Team Leader

Guadalajara, Mexico

Dec 2021 – June 2025

- Founded and led a 50+ student multidisciplinary team across aerodynamics, structures, electrical, and business divisions over four competition cycles, managing timelines, resources, and technical reviews.
- Reduced the vehicle drag coefficient from 0.20 to 0.06 through CFD analysis in ANSYS Fluent, applying NACA four-digit airfoil profiles to key body sections.
- Achieved a 30 kg final vehicle weight via FEA-driven composite design, using tailored layups with a honeycomb core and balsa inserts in stress-concentrating regions.
- Designed and prototyped three steering systems using topology optimization, manufactured via CNC machining and 3D printing (Nylon 6 + CF).
- Manufactured two carbon-fiber monocoques, first by wet layup, later by vacuum resin infusion.

McGill Formula Electric — Formula SAE Electric

Structures Engineer

Montréal, QC

Aug 2024 – Dec 2024

- Designed a monocoque chassis insert (balsa wood / G10) in Siemens NX for weight reduction and increased compressive strength.
- Designed and simulated an underbody downforce device using STAR-CCM+.
- Developed a genetic-algorithm optimization code applying laminate theory to find optimal ply-angle combinations under defined load cases.

SKILLS & INTERESTS

CFD & Simulation: ANSYS Fluent, STAR-CCM+, Luminary Cloud, ANSYS Mechanical, LS-DYNA, Altair

CAD: SolidWorks, Siemens NX, Fusion 360

Programming: MATLAB, Python, Julia

Languages: English (TOEFL C1), French (DELFB2), Spanish (Native)

Certifications: SolidWorks Mechanical Design Professional, SolidWorks Simulation, Flight Mechanics (ISAE-SUPAERO)